Lecture 4. Exploratory Data Analysis

Variation, missing values, covariation, patterns and models

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4.1 What EDA is

Motivation. EDA has no fixed procedure: each answer suggests the next question.

Exploratory data analysis (EDA) is an iterative cycle:

1. generate questions about the data;
2. search for answers by visualising, transforming and modelling;
3. use what you learn to refine the questions or generate new ones.

Early on, investigate every idea. EDA includes **data cleaning** (same tools).

"There are no routine statistical questions, only questionable statistical routines" (D. R. Cox); "Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise" (J. W. Tukey). Two questions are always useful:

- What type of **variation** occurs *within* my variables?
- What type of **covariation** occurs *between* my variables?

Definitions. *Variable*: a measurable quantity, quality or property; *value*: its measured state; *observation*: measurements made under similar conditions (same object, same time); *tabular data*: values, each tied to a variable and an observation.

4.2 Variation

Motivation. The pattern of variation is information; plot the distribution.

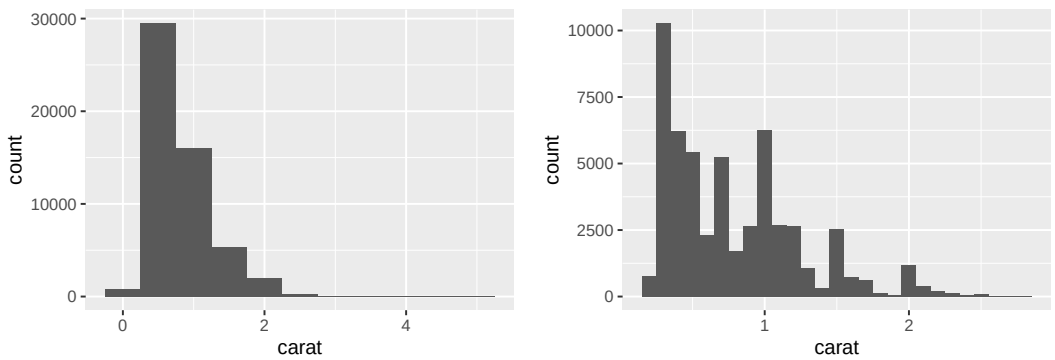
Variation: the tendency of values to change between measurements. The plot depends on the type.

Categorical (few values): bar chart of counts:

```
ggplot(data = diamonds) + geom_bar(mapping = aes(x = cut))
diamonds %>% count(cut)                    # the same numbers
```

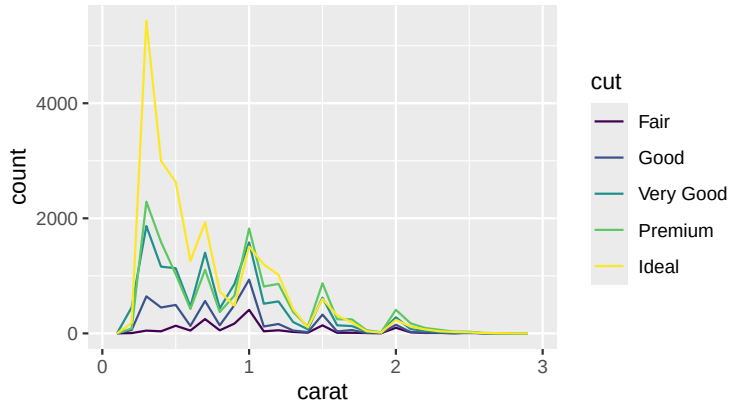
Continuous (numbers, date-times): histogram, counts in equal-width bins:

```
ggplot(data = diamonds) +
  geom_histogram(mapping = aes(x = carat), binwidth = 0.5)
smaller <- diamonds %>% filter(carat < 3)
ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.1)
```



`diamonds %>% count(cut_width(carat, 0.5))` computes the bars. Try several binwidths. `geom_freqpoly()` draws counts as lines, which overlay better:

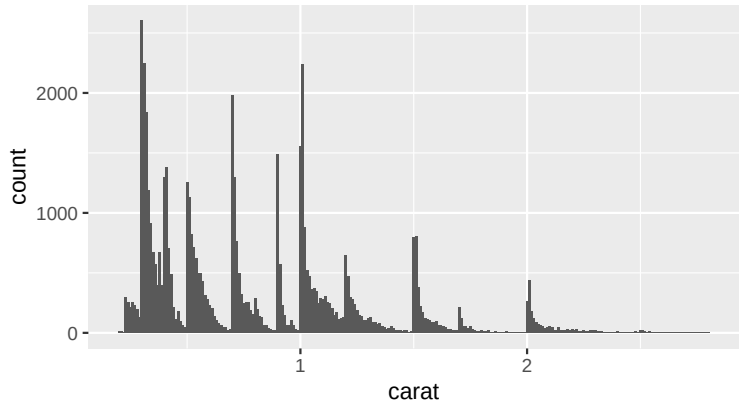
```
ggplot(data = smaller, mapping = aes(x = carat, colour = cut)) +
  geom_freqpoly(binwidth = 0.1)
```



4.2.1 Typical values

Which values are common, which rare, and why? Unusual patterns? Fine binwidth:

```
ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.01)
```

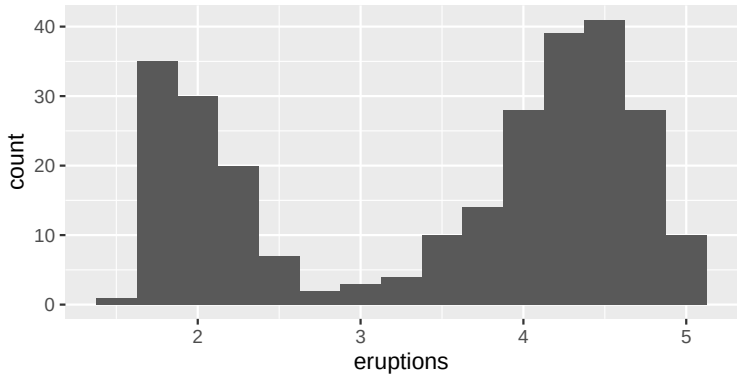


Why peaks at whole carats and common fractions, more mass just right of each peak, nothing above 3 carats? (Cutters aim for round weights, which earn a premium [Supplement].)

4.2.2 Subgroups

Clusters suggest subgroups: how are they alike within, different between, and could they mislead? Old Faithful eruptions are short (about 2 min) or long (4–5 min):

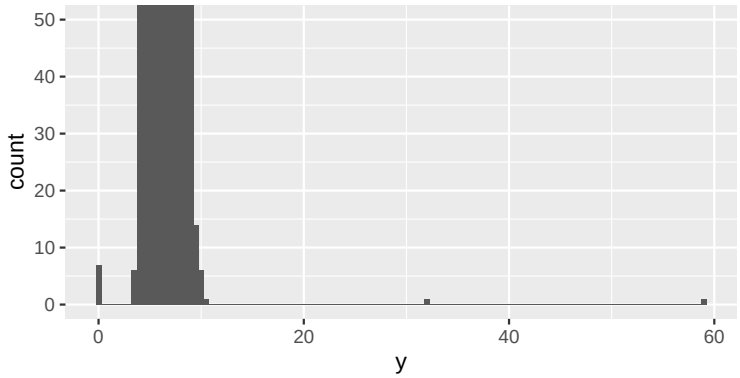
```
ggplot(data = faithful, mapping = aes(x = eruptions)) +
  geom_histogram(binwidth = 0.25)
```



4.2.3 Unusual values

The histogram of width `y` has a suspiciously wide x-axis; `coord_cartesian(ylim = c(0, 50))` (zoom, no data dropped, unlike `ylim()`) reveals values near 0, 30 and 60 mm.

```
ggplot(diamonds) +
  geom_histogram(mapping = aes(x = y), binwidth = 0.5) +
  coord_cartesian(ylim = c(0, 50))
```




```
unusual <- diamonds %>%
  filter(y < 3 | y > 20) %>%
  select(price, x, y, z) %>%
  arrange(y)
unusual
```

```
## # A tibble: 9 x 4
##   price     x     y     z
##   <int> <dbl> <dbl> <dbl>
## 1  5139     0     0     0
## 2  6381     0     0     0
## 3 12800     0     0     0
## 4 15686     0     0     0
## 5 18034     0     0     0
## # i 4 more rows
```

Width 0 is impossible (errors; some have `x`, `y`, `z` all 0); 32 and 59 mm are implausible for stones priced as ordinary. Outliers are data errors or new science:

- repeat the analysis with and without them;
- small effect, no explanation: replace by `NA` and move on;
- substantial effect: find the cause; disclose any removal.

4.3 Missing values

Motivation. Mark bad values missing instead of deleting the rows, which keeps the good measurements.

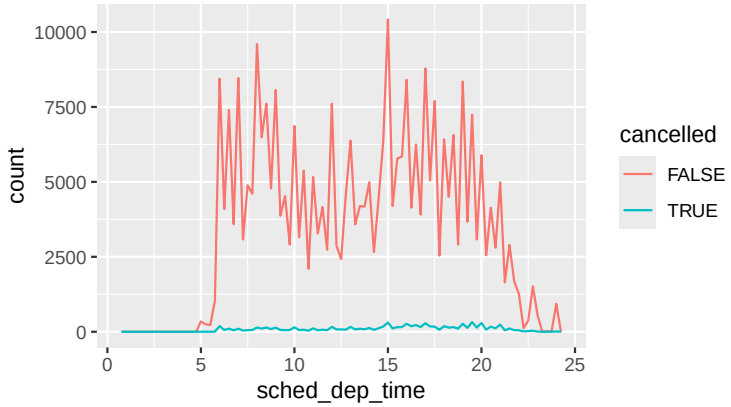
Instead of `filter(between(y, 3, 20))`:

```
diamonds2 <- diamonds %>%
  mutate(y = ifelse(y < 3 | y > 20, NA, y))
```

`ifelse(test, yes, no)` is vectorised: `yes` where `test` is `TRUE`, else `no` (`case_when()` for more cases). ggplot2 drops `NA` with a warning; `na.rm = TRUE` silences it.

Missingness can be informative: missing `dep_time` = cancelled. Scheduled times, cancelled versus not:

```
flights %>%
  mutate(
    cancelled = is.na(dep_time),
    sched_hour = sched_dep_time %/% 100,
    sched_min = sched_dep_time %% 100,
    sched_dep_time = sched_hour + sched_min / 60
  ) %>%
  ggplot(mapping = aes(sched_dep_time)) +
  geom_freqpoly(mapping = aes(colour = cancelled), binwidth = 1/4)
```



Cancelled flights are few, so the comparison is poor (fixed in the next section).

4.4 Covariation

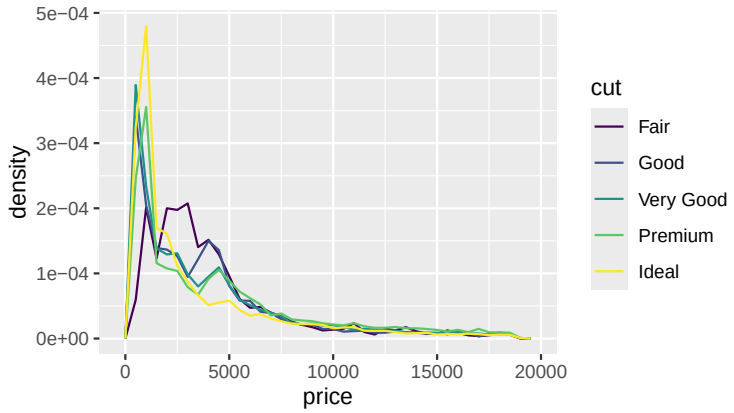
Motivation. How two variables vary together; the plot depends on their types.

Covariation: the tendency of variables to vary together; visualise it.

4.4.1 A categorical and a continuous variable

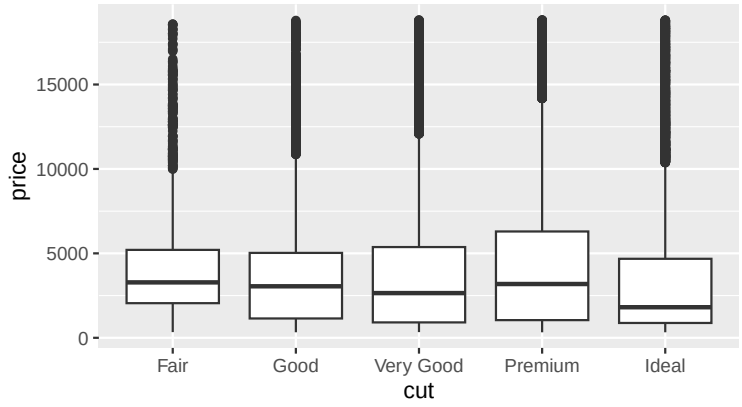
Counts differ greatly between cuts; **density** (area one per polygon) makes shapes comparable:

```
ggplot(data = diamonds, mapping = aes(x = price, y = after_stat(density))) +
  geom_freqpoly(mapping = aes(colour = cut), binwidth = 500)
```



Fair (lowest quality) diamonds seem the most expensive. **Boxplot**: box = IQR with median line, whiskers to the farthest non-outliers, points beyond 1.5×IQR drawn individually.

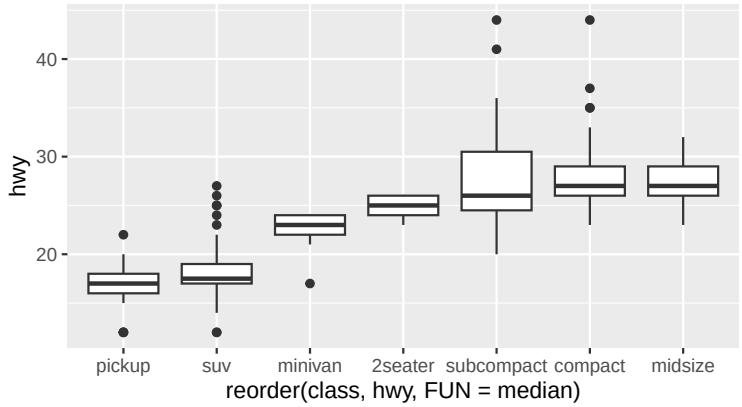
```
ggplot(data = diamonds, mapping = aes(x = cut, y = price)) +
  geom_boxplot()
```



Better diamonds are cheaper on average (explained in Section 4.5).

Reorder unordered categories by a summary:

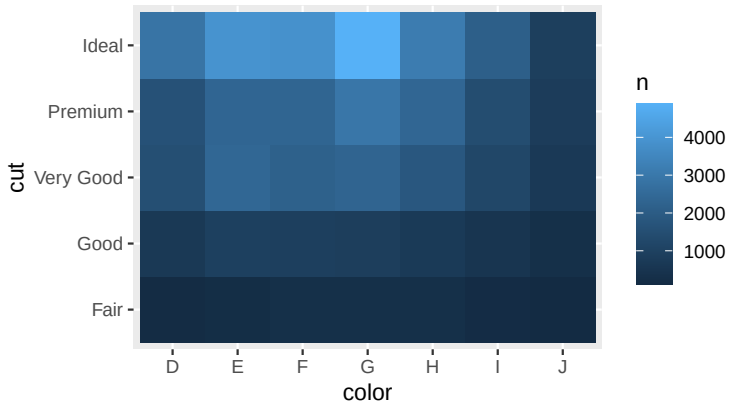
```
ggplot(data = mpg) +
  geom_boxplot(mapping = aes(x = reorder(class, hwy, FUN = median), y = hwy))
```



4.4.2 Two categorical variables

Count each combination: `geom_count()` (circle size ∝ count) or a heat map of dplyr counts:

```
diamonds %>%
  count(color, cut) %>%
  ggplot(mapping = aes(x = color, y = cut)) +
  geom_tile(mapping = aes(fill = n))
```

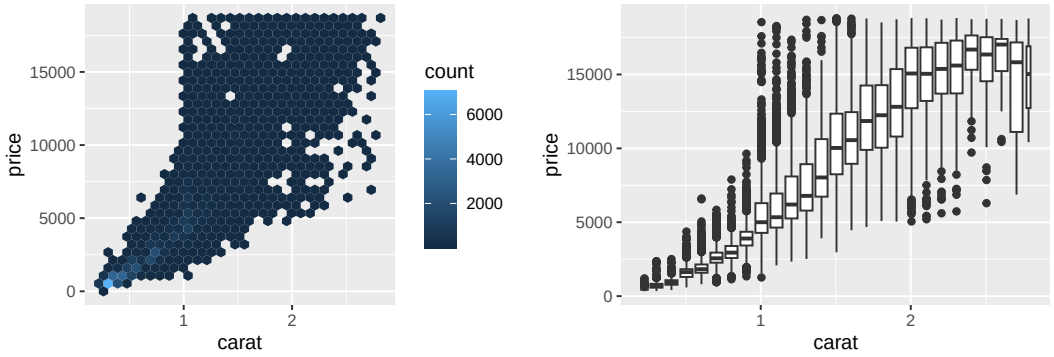


4.4.3 Two continuous variables

54 000 points overplot. Remedies: `alpha = 1/100`; 2-D bins coloured by count, `geom_bin2d()` or `geom_hex()` (needs `hexbin`); or bin one variable to make it categorical:

```
ggplot(data = smaller) +
  geom_hex(mapping = aes(x = carat, y = price))
ggplot(data = smaller, mapping = aes(x = carat, y = price)) +
  geom_boxplot(mapping = aes(group = cut_width(carat, 0.1)))
```

```
## Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous.
## Picking default orientation 'x'.
```



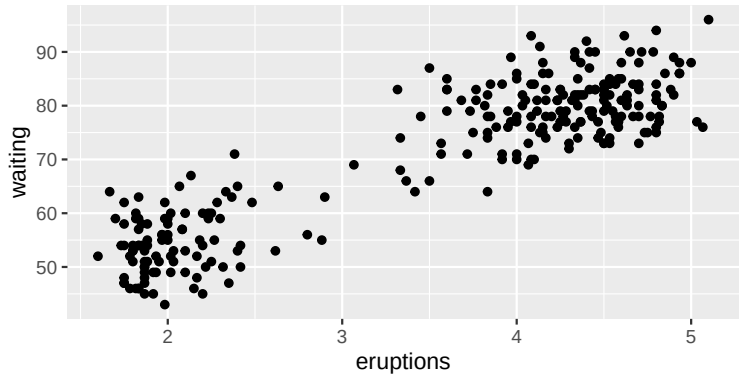
`cut_width(x, w)`: bins of width `w`; `cut_number(x, n)`: `n` bins of about equal size.

4.5 Patterns and models

Motivation. A model removes a strong relationship to reveal weaker ones: the cheap good diamonds.

For a pattern ask: coincidence? How described, how strong? Other variables? Different in subgroups? Old Faithful: longer waits, longer eruptions, two clusters:

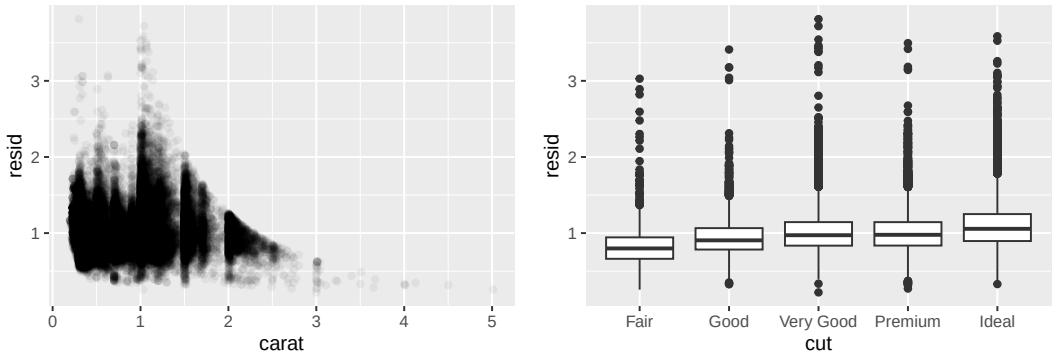
```
ggplot(data = faithful) +
  geom_point(mapping = aes(x = eruptions, y = waiting))
```



Variation creates uncertainty; covariation reduces it (one variable predicts the other). Only causal covariation allows *control*.

Models extract patterns. Price depends strongly on carat, and better cuts are smaller. Fit price on carat and look at the residuals:

```
mod <- lm(log(price) ~ log(carat), data = diamonds)
diamonds2 <- diamonds %>%
  add_residuals(mod) %>%
  mutate(resid = exp(resid))
ggplot(data = diamonds2) +
  geom_point(mapping = aes(x = carat, y = resid), alpha = 1/20)
ggplot(data = diamonds2) +
  geom_boxplot(mapping = aes(x = cut, y = resid))
```



Relative to size, better diamonds are more expensive (models: Lectures 13–14).

Why log–log [Supplement]

$\log(\text{price}) = a_0 + a_1 \log(\text{carat}) \iff \text{price} = e^{a_0}\,\text{carat}^{a_1}$, a power law, linear with roughly constant spread on the log scale. The residual is $\log(\text{price}/\widehat{\text{price}})$, so `exp(resid)` > 1 means dearer than a typical diamond of that size.

```
coef(mod)
```

```
## (Intercept)  log(carat)
##    8.448661    1.675817
```